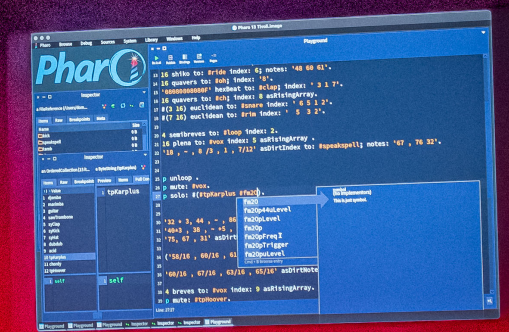


BLOC5

github.com/lucretiomsp/Bloc5



#PCD2021

CASEY
REAS



Thanks Casey, my pleasure

“I think Processing continuing to be used in other programming languages and environments [...], that’s an exciting future, is like breaking the way of thinking out of Processing and Java...”

<https://youtu.be/obzuaPSk3V8?si=kYp3cRIKDBi98QOU>

Inria

A long tradition of teaching code through making

1967

Logo

Seymour Papert's **Turtle graphics**. Code as a way for children to think.

1999

Design By Numbers

John Maeda at M.I.T. Minimal language, visual-first.

2001

Processing

Casey Reas & Ben Fry. Built on Java.

2014

p5.js

Lauren McCarthy. Processing for the web, radically accessible.

Inria

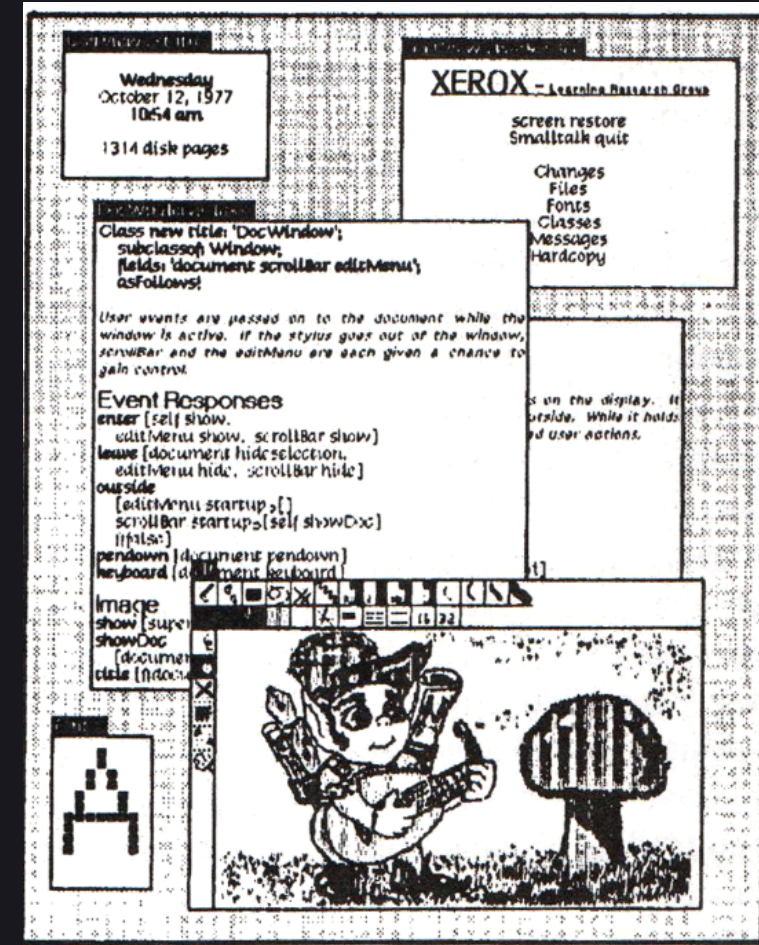
Smalltalk was born (also) for this

Smalltalk was designed at Xerox PARC as a language for **children** : a medium for creativity and learning, not just industry.

The Dynabook, painting interfaces, kids drawing on screens...

Then creativity quietly slipped off the agenda because...

Smalltalk grew up and got a day job...

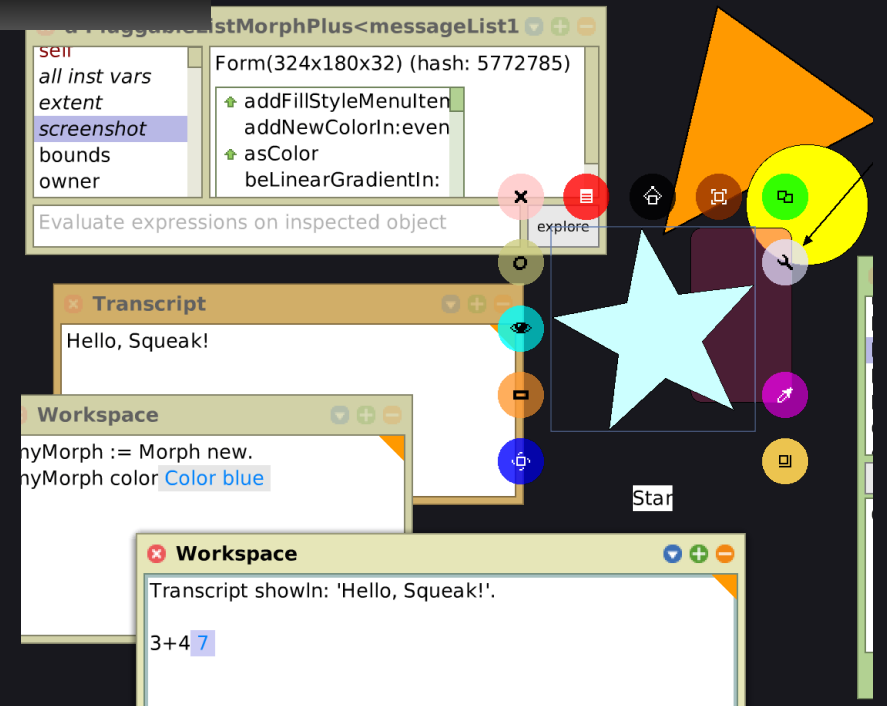


Inria

Morphic for visual art?

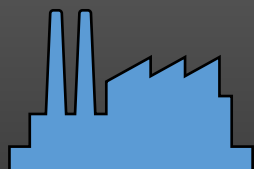
Morphic is **not** what can spark a generation of live coder

```
window := Morph new
  extent: 400@200;
  color: Color white;
  position: 100@100.
circle := (EllipseMorph new
  extent: 100@100; color: Color red; position: 1100@150).
square := (Morph new
  extent: 100@100; color: Color blue; position: 250@150).
triangle := (PolygonMorph vertices: { 50@0. 100@100. 0@100 }
  color: Color green
  borderWidth: 2
  borderColor: Color black).
window addMorph: circle; addMorph: square; addMorph: square.
window openInWorld.
window delete.
```



Inria

Pharo today



Industry

Widespread for business applications, trusted by many demanding commercial clients.



Research

A great platform for language and research.



Creativity?

Almost nothing until music arrived in 2022.

lnria

First, sound. Now, vision.



COYPU (2022) :

Programming music on-the-fly with Pharo.

PHAUSTO (2024) :

sound synthesis (embedding Faust) within Pharo.

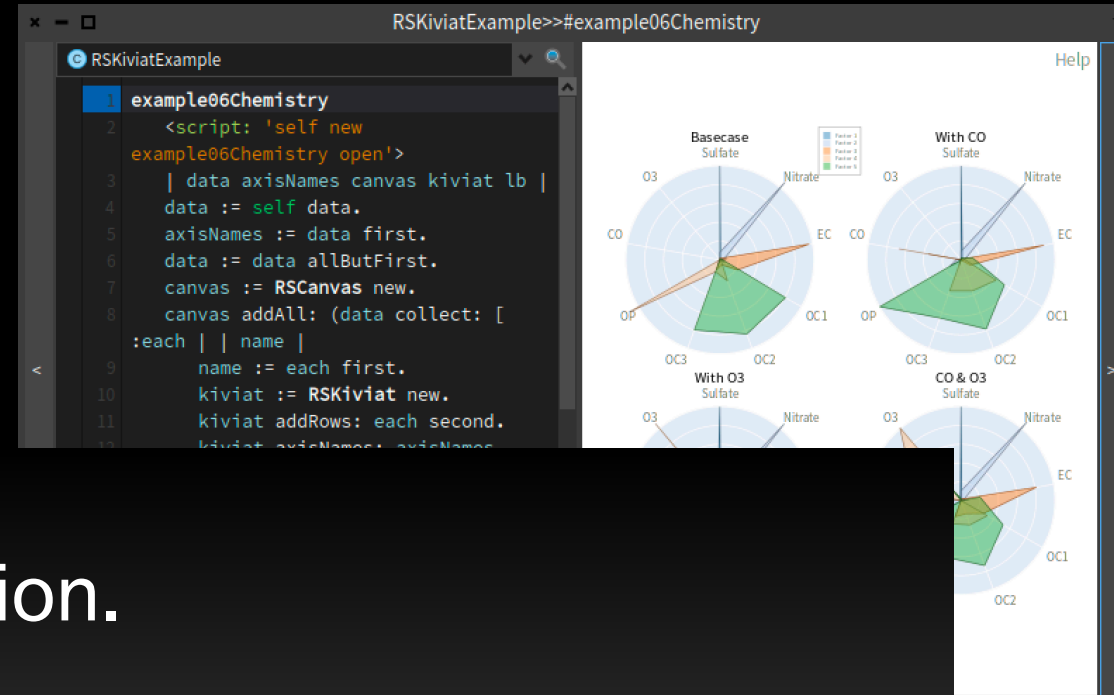


Pharo can now make **sound**. It was missing **visual art**.

Inria

But doesn't Roassal already draw?

Roassal is excellent , for **data visualization**.



Charts, graphs, trees, maps of data.

→ Bind data to: produce a visualization.

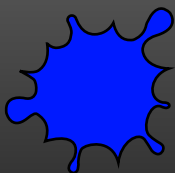
But Bloc5 IS MADE FOR SKETCHING.

→ A Different goal: free creative expression, not insight from datasets.

Inria

Bloc5, a p5js inspired API for Bloc

A friendly layer that turns Bloc's retained-mode scene graph into an immediate-mode sketching experience, while still taking advantage of many of its features.



Artists

Express ideas in code
without fighting the API.



Kids

A gentle, visual first
contact with programming.



Beginners

No coding literacy
required.

What is Processing / p5.js?

- p5.js is a friendly tool for learning to code and make art. It is a free and open-source JavaScript library built by an inclusive, nurturing community.
- You write a little program and visual arts appear immediately.
- Bare calls like: `fill`, `circle`, `square` are implicitly methods on a hidden global sketch object. p5 just omits the receiver for convenience.

```
function setup() {  
  createCanvas(400, 400);  
}  
function draw() {  
  background(0);  
  circle(mouseX, mouseY, 50);  
}
```

Two main functions:

setup() runs once /

draw() runs on every frame.

What is Bloc?



Bloc is a low-level UI infrastructure & framework for Pharo.
State of the art library, but the API is verbose for sketching.

DAY-02
WED. JULY 8ND

🕒 3:00 pm - 3:30 pm

Bloc for Pharo: What's New?

Martin Dias and Pablo Tesone

Main Amphi

Inria

An economic API

18181E

ABSTRACTION! ABSTRACTION! ABSTRACTION! ABSTRACTION!
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Bloc



BLOC5

```
line := BIElement new  
  geometry: (BILineGeometry new from 20@20 to: 300 @ 100;  
  border: (BIBorder paint: Color green width: 8)
```

```
line := B5Line new from 20@20 to: 300 @ 300;  
  stroke: Color green  
  strokeWidth: 8
```

Inria

BLOC5's syntax fits on a Rizla

```
square := B5Square new
    extent: 100@100; fill: Color blue; position: 10@10 mode: #corner.
triangle := B5Triangle new v1: 50@0; v2: 100@100; v3 0@100; fill: Color green.
circle := B5Circle new diameter: 30; position: 300@300; fill: Color red.
line := B5Line new from 20@20 to: 300 @ 300; stroke: Color green; strokeWeight: 8.
txt := B5Text new text: 'ameega'; fontName: 'Helvetica'; fontSize: 44; position:
200@200.
cat := B5Image new fromFile: 'images/chuckCat.png'; extent: 200@200.
circle onKeyDown: [ :key | circle background: Color random].
square onMouseDown: [ :x :y | square position: x@y ].
triangle translate: 100@100.
cat rotate: 23.
space := B5Space new frameRate: 144 bpmAsFrameRate.
space drawAnimation: [ circle position: 300@(Random new * 400) ]
```

GOMME ARABIQUE NATURELLE

inria

BLOC

VS

BLOC5

```
square := BIElement new
  size: 100@100;
  geometry: BIRectangleGeometry new;
  background: Color blue.

triangle := BIElement new
  size: 100@100;
  geometry: (BIPolygonGeometry vertices: { 50@0. 100@100. 0@100 });
  background: Color green.

circle := BIElement new
  size: 30@30;
  geometry: BICircleGeometry new;
  background: Color red.

space := BISpace new.
space root addChild: square; addChild: triangle; addChild: circle
space show.
```

```
square := B5Square new
  extent: 100@100; fill: Color blue.

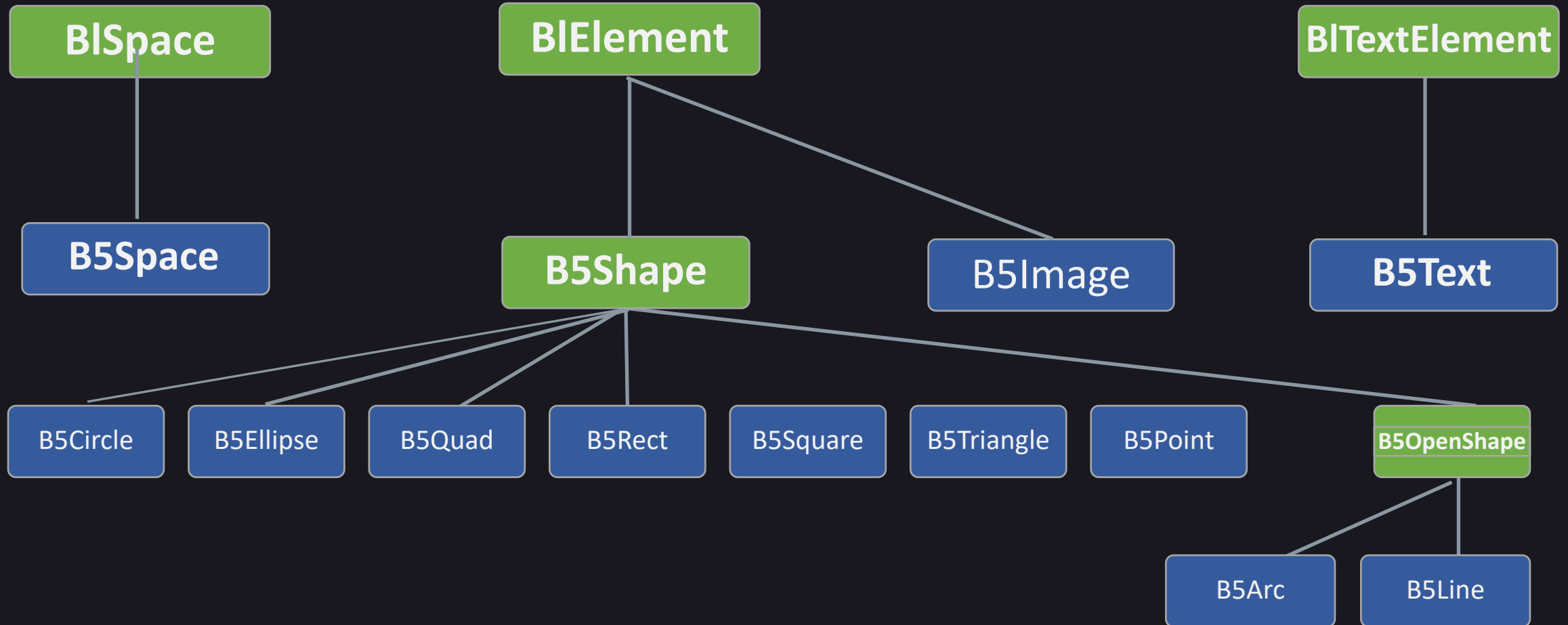
triangle := B5Triangle new
  v1: 50@0; v2: 100@100; v3: 0@100;
  fill: Color green.

circle := B5Circle new
  diameter: 30; fill: Color red.

space := B5Space new.
space root addChild: square; addChild: triangle; addChild: circle
space show.
```

Inria

BLOC5 class hierarchy



Animation in one message

drawAnimation: corresponds to p5's *draw()*:
the block runs every frame.

```
space := B5Space new frameRate: 30.  
c := B5Circle new position: 200@200; diameter: 80; fill: Color red.  
space addChild: c.  
t := 0.  
space drawAnimation: [  
    t := t + 0.05.  
    c position: 200 @ (200 + (50 * t sin)) ].  
space show.
```

drawAnimation: aBlock

```
| task |  
task := BlRepeatedTaskAction new  
    delay: (1000 / frameRate) milliseconds;  
    action: aBlock;  
    yourself.  
self enqueueTask: task
```

Interaction: event traits

A B5EventHandler trait gives every shape p5-style event hooks.

Mouse

```
onClick:  onMouseDown:  onMouseUp:  
onMouseEnter:  onMouseLeave:  onMouseWheel:
```

Keyboard

```
onKeyDown:  onKeyUp:
```

```
shape onClick: [ :x :y | shape fill: Color random ].  
shape onKeyDown: [ :key | shape position: shape position + (10@23) ].
```

Drawing Order

- p5.js: paint order = call order (immediate mode, irreversible)
- Bloc5: shapes are real children in Bloc's scene graph.
- Parent-child structure lets you transform whole "families" together rotate/translate/scale a group as one unit
- `addChild`: exposed deliberately -> no p5 equivalent

System variables: mouseX / mouseY

Live cursor position, exactly like p5.js.

mouseX / mouseY : current frame.

pmouseX / pmouseY : previous frame.

```
space drawAnimation: [  
  c position:  
    space mouseX @ space mouseY  
].
```

Mapping and normalising values

```
space mouseX map: # (0 600 200 -300) .
```

```
space mouseY norm: # (0 600)
```

Bloc5 + Phausto = Audio/Visual Experience

Graphics from Bloc5. Sound from Phausto. One live image.

Pharo becomes a full creative-coding environment:
audio and visual, live, in one place.

B5

Bloc5

+



Phausto

=



A/V

Inria

Future work



Full p5.js coverage

Bezier Curves and splines, Color Effects, Full Typography

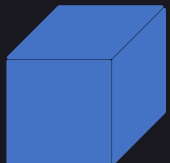


Full tutorials and examples



Fixed background

At the moment, user cant choose to redraw the background



3D rendering

With raylib via Pharo FFI bindings would be the best option.

Conclusions

Bloc5 is good for Pharo

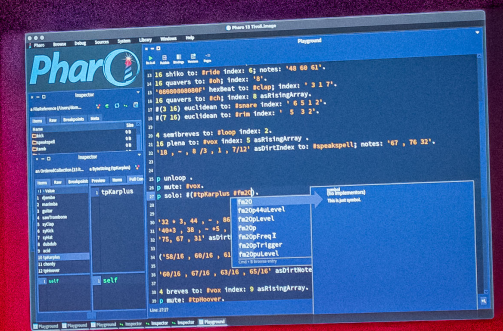
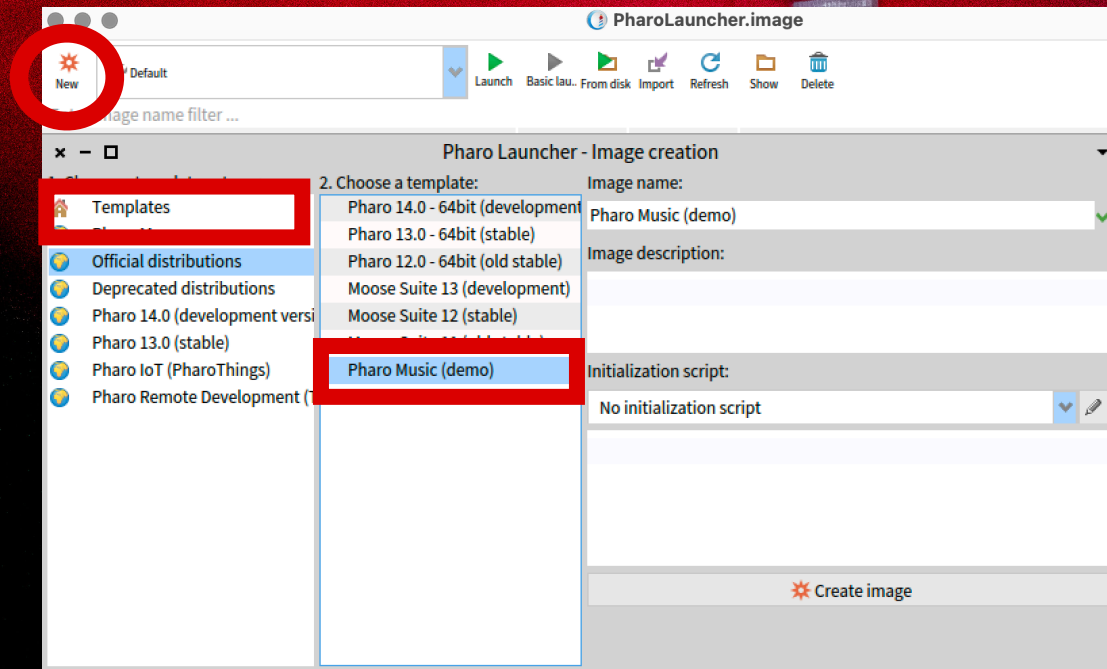
- Immediate visual feedback for kids and non-programmers
- Can bring creative coders to Pharo and the Smalltalk world
- Real-time stress-and-test for underlying Bloc infrastructure

What to improve

- Documentation, examples, tutorials are fundamental!
- Explicit choices on idiomatic choices
- Better handling for exceptions (debugger explosions on events)

<https://github.com/lucretiomsp/Bloc5>

A Pharo Music Image with Coypu, Phausto, CoypuIDE, Bloc5.
For artists, for kids, for punks, for greybeards. For everyone.



Thank you, have fun and contribute!



<https://github.com/lucretiomsp/Bloc5>

Íria